

DATA EXCEL PROJECT

Project Objective:

This project focuses on cleaning, transforming, and analyzing retail sales data using Microsoft Excel. The objective was to convert raw transactional data into meaningful business insights through data preparation, pivot table analysis, KPI tracking, and data visualization. The project began with data cleaning activities such as standardizing date formats, handling missing values, splitting customer names into separate fields, and ensuring data consistency across multiple datasets. A structured data model was created using fact and dimension tables to improve data organization and analysis. Descriptive analysis was performed using Pivot Tables to identify key business trends, including top-performing products, highest-value customers, monthly sales performance, payment method preferences, and category-wise sales distribution. Key Performance Indicators (KPIs) such as Total Sales, Total Profit, Total Quantity Sold, and Total Customers were calculated to measure overall business performance. Diagnostic analysis was also conducted to investigate factors affecting sales and profitability, such as the impact of discounts on revenue generation and product performance. Interactive charts and dashboard elements were created to present findings in a clear and visually appealing format. The project demonstrates practical Excel skills in data cleaning, data transformation, Pivot Tables, KPI reporting, dashboard creation, and business analysis, making it a valuable end-to-end retail sales analytics project.

Dataset Description:

Data contains of

1. Sales ID
2. Customer Id
3. Product id
4. Customer name
5. Order date
6. Quantity
7. Units sold
8. Discount
9. Total amount
10. Payment type.

Column Description

Column Name	Description
Sales id	Id for each sale
Customer id	Id for each customer
Product id	Id for each customer
Quantity	No of products
Total Amount	Total Sales Amount
Order Date	In which date product been ordered
Customer Name	Name of an employee
Units sold	How many units been sold
Discount	Some percentage of amount will be deducted from the payment amount
Payment Type	Type of payment customer done most

- Data cleaning & transformation:

Checked all worksheets for duplicate records, while no duplicates were found in Sheets 1, 2, 3 and 4.

- Applied the RIGHT function to extract the numerical portion of the Customer ID field.
- Split the Customer Name column into First Name and Last Name using the delimiter feature.
- Utilized the PROPER, TRIM, and CLEAN functions to standardize text formatting, remove extra spaces, eliminate non-printable characters, and ensure data consistency.
- Used the Find and Replace feature to correct misspelled words and improve data accuracy.
- Applied the IF function to replace missing values with "UNKNOWN" or "N/A", depending on the context of the data.
- Converted the Order Date column to the DD-MM-YYYY format using the TEXT() function to ensure consistency in date representation across the dataset.
- Applied the IF(ISBLANK()) function to identify missing values in the Sales Fact table and replaced them with "UNKNOWN" where the required information was unavailable.

Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

- Descriptive analysis:

What are the monthly sales trends?

Who are the top customers?

Which products generate the highest profit?

Which payment method used most?

- Diagnostic Analysis:

Why do some categories have high sales but low profit?

Why are some regions underperforming?

Prescriptive Analysis:

Reduce discounts on low-margin products.

Review pricing strategies for top-selling products.

Focus promotions on products with higher profit margins.

Increase marketing for underperforming products.

Bundle low-selling products with popular items.

If a payment method generates most revenue:

Offer incentives for preferred payment methods.

If certain months have low sales:

Launch seasonal promotions during slow periods.

Conclusion:

The project successfully transformed raw retail sales data into meaningful insights using Excel. It improved data quality through cleaning and enabled effective analysis using Pivot Tables, KPIs, and dashboards. The results support better decision-making by highlighting sales trends, profitability drivers, and key business performance metrics.